



CAUSAL INFERENCE | RESEARCH PRESENTATION

Instant payments and labor formalization

Evidence from PIX in Brazil

A triple-difference analysis of 5,570 municipalities, 88 sectors, 2017-2023

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Motivation

Brazil bet on instant payments as Digital Public Infrastructure. The labor market effect is theoretically ambiguous.



93%

of Brazilian adults use PIX, four years after launch (Nov 2020)



38%

of Brazil's labor force works informally — a top policy priority



?

Did PIX expand or substitute formal employment?

Sources: BCB (PIX adoption); IBGE/PNAD-C (informality); Wikipedia.

Research Question

Did the launch of **PIX** in November 2020 **expand or substitute** formal employment in Brazil?

TWO COMPETING HYPOTHESES

H1: COMPLEMENTS



Digital trail forces formalization

PIX leaves a record of transactions. To use it as a business, firms must register, pay taxes, and hire formally. → Formal employment grows.

H2: SUBSTITUTES



Lower cost of being informal

PIX lets autonomous workers and informal firms transact digitally without registering. → Formal employment falls (Ulyssea 2018, AER).

This paper lets the data adjudicate.

Where this study fits

EXISTING CAUSAL EVIDENCE ON PIX

Burga (2024, IDB WP)

PIX raised wages in cash-intensive sectors. Intensive margin only.

Sarkisyan (2024)

PIX boosted bank deposits, ~US\$380/quarter consumer welfare gain.

Leal & Haase (2025)

PIX reduced bank branches by 0.25 per municipality.

Ulyssea (2018, AER)

Two margins of informality in Brazil: 40% of informal workers are in formal firms.

THE GAP

Existing causal evidence focuses on the **intensive margin** (wages within formal jobs, deposits within bank accounts).

The **extensive margin** — whether PIX created or destroyed formal jobs in absolute terms — is unanswered.

This question matters most for understanding informality dynamics and the welfare implications of DPI.

Methodology

Triple-difference (DDD): mobile penetration × cash-intensive sectors × post-PIX

$$Y_{mst} = \beta \cdot (\text{Mobile}_m \times \text{Cash}_s \times \text{Post}_t) + \alpha_{mt} + \gamma_{st} + \delta_{ms} + \varepsilon$$



Mobile_m

ANATEL mobile penetration in 2019,
pre-PIX (Burga 2024)



Cash_s

Cash-intensive CNAE sectors (retail,
food, services)



Post_t

= 1 if year >= 2021 (PIX launched Nov
2020)



FE

Mun*Year, Sector*Year, Mun*Sector
— absorb COVID, Auxilio

Identifying assumption: *Cash-intensive sectors in high-mobile munis would have evolved like other sectors absent PIX (testable via event study).*

Why a triple-difference design?

PIX is a national shock with no temporal variation across municipalities. Standard DiD and its modern extensions don't apply.

	Simple DiD	PIX is a national shock — all munis treated simultaneously. No control group.
	Staggered DiD (Callaway-Sant'Anna)	Requires staggered rollout across units. PIX launched once, nationwide.
	Synthetic control	Designed for a single treated unit (e.g., one country). Brazil treated entirely.
	Triple-difference (DDD)	Exploits three sources of variation: geographic + sectoral + temporal.

DDD is the standard approach for national shocks with cross-sectional heterogeneity (e.g., Burga 2024).

How DDD handles the pandemic confounder



The challenge

PIX launched in Nov 2020, in the middle of the pandemic. Auxilio Emergencial transferred BRL 300+ bn to ~70M Brazilians, requiring digital banking. *A simple before/after design would conflate PIX, COVID, and Auxilio.*



How DDD solves it

Municipality * Year FE absorb **any shock** hitting all sectors equally in a given municipality and year — including COVID, Auxilio, local recessions. β is identified from **differential responses across sectors within the same municipality-year**. The pandemic affected all sectors; PIX disproportionately affected cash-intensive ones.

Why sectoral heterogeneity is the test

Devil's advocate: "Maybe more mobile = more development = different employment dynamics?"

IF IT WERE "GENERAL DEVELOPMENT"

Effect would appear **uniformly across all sectors** in high-mobile municipalities:

- ✗ Retail (cash-heavy) → changes
- ✗ Bars (cash-heavy) → changes
- ✗ Manufacturing → changes
- ✗ Construction → changes

Effect everywhere = it's not PIX, it's just development.

IF IT IS A PIX EFFECT

Effect should appear **only in cash-intensive sectors** — those PIX directly disrupts:

- ✓ Retail (cash-heavy) → affected
- ✓ Bars (cash-heavy) → affected
- Manufacturing → flat
- Construction → flat

Sectoral signature = the fingerprint of PIX.

The triple interaction (mobile × cash × post) isolates the second pattern from the first. The heterogeneity IS the test.

Data

*Municipality * Sector * Year panel, 2017-2023, all data via basedosdados.org*



Outcome

Log(formal employment relationships) by municipality * CNAE 2-digit

RAIS (Relacao Anual de Informacoes Sociais), Ministry of Economy



Treatment exposure

Mobile access (4G+) per capita in 2019 — pre-PIX, exogenous (Burga 2024)

ANATEL (Brazilian telecom regulator)



Sector exposure

Cash-intensive sectors: retail, food, accommodation, transport, repair, services

Manual classification following Burga (2024); 11 CNAE sections



Controls

Municipal population, GDP per capita

IBGE (national statistics agency)

Sample at a glance

After cleaning and merging, the analysis panel covers:

5,570

Brazilian municipalities

88

CNAE 2-digit sectors

7

years (2017-2023)

1.17M

municipality-sector-year obs

[INSERT FIGURE — figures/01_raw_trends.png]
Mean log employment by sector type and mobile penetration quartile

Main results — Triple-DiD estimates

	(1) Baseline	(2) State FE	(3) Drop 2020-21	(4) Two-way cluster
Mobile × Cash × Post	−0.001*	−0.005***	−0.002**	−0.003
	(0.001)	(0.002)	(0.001)	(0.002)
Observations	1,167,578	1,167,585	831,727	1,167,578
R² Adj.	0.942	0.941	0.935	0.942
Mun × Year FE	Yes	—	Yes	Yes
Sector × Year FE	Yes	Yes	Yes	Yes
Mun × Sector FE	Yes	Yes	Yes	Yes

HEADLINE

−0.5%

relative decline in formal employment for cash-intensive sectors per 1-SD increase in mobile penetration, post-PIX (state FE spec.)

*Effect: opposite to formalization hypothesis (H1).
Consistent with substitution (H2).*

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. SEs clustered by municipality. (4) cluster by mun + sector.

Parallel trends — the identification check

Year-by-year coefficients of triple interaction (ref: 2019)

[INSERT FIGURE — figures/04_event_study.png]

READING THE PLOT

PRE-PIX (2017-2019)

Coefficients near zero, statistically insignificant.
→ *parallel trends holds.* ✓

POST-PIX (2020+)

Negative coefficients emerging, peaking in 2021.
→ *effect is concentrated in 2021-2022.*

Dynamic effects — the effect is transitory

Separate coefficient for each post-PIX year. The effect peaks in 2021 and reverts by 2023.



Interpretation: PIX generated a transitory shock (~2 years) in cash-intensive sectors of high-mobile munis. The market adjusts: by 2023, formal employment in these cells is statistically indistinguishable from the rest. Consistent with a 'shake-out' equilibrium where some workers shifted to digital informality and the labor market re-equilibrated.

Robustness

The result survives a battery of sensitivity checks



Placebo (fake post = 2018)

Coefficient: -0.0004 (n.s.).
Pre-trends valid. ✓



Drop 2020-21 (pandemic)

Coefficient: -0.002^{**} (still sig).
Effect not solely COVID.



State-level FE (conservative)

Coefficient: -0.005^{***} (stronger).
Result robust.



Two-way clustering

Magnitude unchanged (-0.003). $p = 0.17$. Reported.

What we learned



PIX did not expand formal employment — it relatively reduced it

In municipalities with high mobile exposure to PIX, cash-intensive sectors saw RELATIVELY slower formal employment growth post-PIX. Hypothesis H1 (formalization) is rejected; H2 (substitution) gains support.



The effect is transitory, not permanent

Peak in 2021 (-0.005^{***}), attenuating in 2022 (-0.003^{**}), and fully reverted by 2023 (n.s.). Consistent with a 'digital informality shake-out': workers shift to digital autonomy, then labor markets re-equilibrate.



Theoretical anchor: Ulyssea (2018, AER)

Brazil has two informality margins. ~40% of informal workers are inside formal firms. Lowering the cost of operating informally (via digital payments) substitutes formal labor relations — at least temporarily.

Limitations & open questions

Honest acknowledgement of what the design cannot answer

Mobile as proxy for PIX exposure

Treatment exposure inferred from 2019 mobile data. Direct PIX adoption microdata not yet public. Following Burga (2024) approach.

Cash-intensity classification

Sector-level proxy. Some heterogeneity within sectors not captured. Manual classification follows literature (11 CNAE sections).

Cannot directly observe substitution

RAIS captures only formal employment. Informal sector data (PNAD-C) does not have municipal granularity. Substitution is inferred, not measured.

External validity

Brazil-specific institutional setting (banks mandated to offer PIX, zero P2P fees). Mechanism may differ in other DPI deployments.

Bibliography

Burga, C. (2024). *Financial Technologies, Labor Markets, and Wage Inequality: Evidence from Instant Payment Systems*. IDB Working Paper.

Ulyssea, G. (2018). *Firms, Informality, and Development: Theory and Evidence from Brazil*. *American Economic Review*, 108(8): 2015–47.

Sarkisyan, S. (2024). *Instant Payment Systems and Bank Competition*. Working paper.

Leal, R., & Haase, B. (2025). *Instant Payments and Banks*. *Economia Ensaaios*.

Ponczek, V., & Ulyssea, G. (2022). *Enforcement of Labour Regulation and the Labour Market Effects of Trade*. *The Economic Journal*, 132(641): 361–390.

VoxDevLit (2023). *Informality*. *VoxDev Literature Review*, Volume 6.

International Monetary Fund (2023). *Pix: Brazil's Successful Instant Payment System*. *IMF Country Report* 2023/289.

Methodological references: Goodman-Bacon (2021), de Chaisemartin & D'Haultfœuille (2020), Olken (2007 AER) on triple-difference designs.



TAKEAWAY

Digital payment infrastructure can substitute, not just complement, formal employment.

Mechanism: By lowering transaction costs, PIX enabled cash-intensive activity to operate outside the formal labor market — at least transitorily. The effect peaks in 2021, attenuates in 2022, and reverts by 2023.

Policy implication: DPI design should consider its interaction with informality regulation. Efficiency gains in payments may shift activity into informal arrangements unless complemented by formalization incentives.

Thank you. | Questions?